WAEL MELKI

DATA SCIENTIST | DATA ANALYST | AI / ML ENGINEER

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PROFESSIONAL PROFILE

ICT engineer from **SUP'COM** (graduated with **Très bien**) specializing in the intersection of **data science, applied machine learning, AI engineering, business analytics and software development**. Built end-to-end systems spanning multi-source data integration, SQL analytics, explainable predictive models, BI dashboards, MLOps and RF simulation. At Huawei, developed a 5G planning decision pipeline covering **12,412 subscribers and 731 radio cells**; combines quantitative rigor with software engineering and domain-aware decision making.

PROFESSIONAL EXPERIENCE

Huawei North Africa Feb 2026 – Jul 2026
Final-Year Engineering Intern – AI-Driven 5G Fixed Wireless Access Planning Tunis, Tunisia

- Architected a **Python/PostgreSQL** decision pipeline consolidating CRM, customer usage, service consumption, coverage and RAN KPIs across **5 data sources**; assembled a reliable **50-column** analytical dataset for **12,412 subscribers** and **731 radio cells**.
- Implemented data profiling, deduplication, outlier checks and referential integrity controls; identified and resolved systematic quality anomalies before feature engineering and model training.
- Built **explainable FWA eligibility scoring**, **K-Prototypes** customer segmentation and a calibrated **XGBoost** churn classifier (held-out ROC-AUC ~**0.71**) to drive offer recommendations and customer prioritization.
- Modeled radio-network feasibility using **PRB-based capacity simulation**; identified **2,629 eligible subscribers** and **658 priority contacts**, with projected peak load **68.26%**, below the **80%** operational ceiling.
- Produced an actionable, database-backed campaign shortlist and decision-support outputs joining commercial value, churn risk and engineering constraints. *Stack: Python, SQL, Pandas, scikit-learn, XGBoost, PostgreSQL, Power BI, Git.*

NetData Innovation Jul 2025 – Aug 2025
Engineering Intern – 3D RF Coverage Simulation and Performance Optimization Remote, France

- Developed a Python/C++ desktop simulation application for **Wi-Fi/5G radio coverage** in 3D scenes, combining configurable antennas, propagation analysis and interactive **PyQt6 / PyVista / VTK** visualization.
- Profiled and optimized computational workloads using **Octree spatial structures**, multiprocessing and C++/Python refactoring; **reduced processing time by 70%** in reported tests.

Tunisie Telecom First year
Human Development Internship – Stage de formation humaine Tunisia

TECHNICAL TOOLKIT

Programming & analytics: Python, SQL, C++, MATLAB; Pandas, NumPy, data cleaning, EDA, ETL, feature engineering, data validation.

AI & ML: scikit-learn, XGBoost, PyTorch, K-Prototypes, Random Forest, SVM, Isolation Forest, HDBSCAN, SHAP; classification, clustering, anomaly detection, model evaluation.

Data & BI: PostgreSQL, SQL Server, relational modeling, star schema, Power Query, Power BI, DAX, dashboards, KPI reporting.

ML engineering: MLflow, ZenML, DVC, Evidently, MinIO, Docker / Docker Compose, Git, Linux, reproducible pipelines, drift monitoring.

Simulation & signals: 5G FWA, LTE, RAN KPI, PRB capacity, RF propagation, signal/image processing, OpenCV, VTK, PyVista, IoT.

SELECTED ENGINEERING & DATA PROJECTS

CIFAR-10 MLOps Training and Monitoring Platform View repository
Designed reproducible computer-vision training workflows orchestrated with **ZenML** and **MLflow** for experiment tracking and model registry. Versioned datasets with **DVC**, used **MinIO / PostgreSQL** for artifact and metadata storage, and containerized services with **Docker Compose**; integrated **Evidently** drift checks and guarded retraining.

Energy Time-Series Analytics and Anomaly Detection View repository
Collaborated on a signal-processing and unsupervised-learning pipeline for energy-consumption analysis. Applied **coif5 wavelet** compression, detected unusual usage profiles with **Isolation Forest / HDBSCAN**, and exposed interactive results through a **Streamlit** analytical interface.

AdventureWorks Sales Analytics – Power BI View repository
Created an analytics workflow joining **SQL Server** and CSV inputs, **Power Query** transformation, a dimensional **star schema** and **DAX** measures. Built interactive reporting to analyze sales revenue, margins, growth and year-over-year KPIs for business stakeholders.

Non-Invasive Diabetes Detection – Multimodal ML Academic research project
Co-developed a research prototype using **ECG and respiratory signals**: frequency filtering, peak detection, heart-rate variability and breathing-rate feature extraction. Evaluated **SVM / Random Forest** models with subject-aware validation and combined their probabilities using late fusion; small-sample research, not a clinical diagnostic system.

Real-Time Object Tracking – Computer Vision Academic project
Implemented and compared object-tracking algorithms in **Python / OpenCV**, examining performance in dynamic scenes and applying image-processing and algorithmic evaluation techniques.

RF Design and Connected-IoT Prototypes Academic projects
Designed and simulated a **T-network RF attenuator** in **ADS**. Developed an **ESP32 / Arduino** fire/incident detection prototype that triggers automatic **ntfy** notifications, combining sensing, embedded software and connected services.

Statistical Voice Conversion – Digital Signal Processing Academic project
Transformed speech characteristics through **spectral-envelope estimation and modification** with **Python, NumPy and SciPy**, applying statistical analysis and audio signal processing.

EDUCATION

SUP'COM – Higher School of Communications of Tunis 2023 – Sep 2026
National Engineering Diploma in ICT | Specialization: Smart Image Applications. **Graduated with Très bien**; final-year project **16/20**. Focus: AI/ML, mobile networks, signal and image processing, cloud, Big Data.

IPEIT – Preparatory Institute for Engineering Studies of Tunis 2021 – 2023
Preparatory cycle in Mathematics and Physics; ranked **63rd / 920 (top 7%)** in the national engineering entrance examination.

CERTIFICATIONS & ADDITIONAL INFORMATION

Training: MathWorks Machine Learning Onramp, MATLAB Onramp, Stateflow Onramp.

Languages: Arabic (native), French (fluent), English (fluent).

Competitive programming: **92** problems solved on Codeforces (algorithmic problem solving).